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GFAP immunoreactivity within the rat nucleus ambiguus after laryngeal nerve injury

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Abstract

Changes that occur in astroglial populations of the nucleus ambiguus after recurrent (RLN) or superior (SLN) laryngeal nerve injury have hitherto not been fully characterised. In the present study, rat RLN and SLN were lesioned. After 3, 7, 14, 28 or 56 days of survival, the nucleus ambiguus was investigated by means of glial fibrillary acidic protein (GFAP) immunofluorescence or a combination of GFAP immunofluorescence and the application of retrograde tracers. GFAP immunoreactivity was significantly increased 3 days after RLN resection and it remained significantly elevated until after 28 days post injury (dpi). By 56 dpi it had returned to basal levels. In contrast, following RLN transection with repair, GFAP immunoreactivity was significantly elevated at 7 dpi and remained significantly elevated until 14 dpi. It had returned to basal levels by 28 dpi. Topographical analysis of the distribution of GFAP immunoreactivity revealed that after RLN injury, GFAP immunoreactivity was increased beyond the area of the nucleus ambiguus within which RLN motor neuron somata were located. GFAP immunoreactivity was also observed in the vicinity of neuronal somata that project into the uninjured SLN. Similarly, lesion of the SLN resulted in increased GFAP immunoreactivity around the neuronal somata projecting into it and also in the vicinity of the motor neuron somata projecting into the RLN. The increase in GFAP immunoreactivity outside of the region containing the motor neurons projecting into the injured nerve, may reflect the onset of a regenerative process attempting to compensate for impairment of one of the laryngeal nerves and may occur because of the dual innervation of the posterior cricoarytenoid muscle. This dual innervation of a very specialised muscle could provide a useful model system for studying the molecular mechanisms underlying axonal regeneration process and the results of the current study could provide the basis for studies into functional regeneration following laryngeal nerve injury, with subsequent application to humans.

Key words: astrocytes; glial fibrillary acidic protein; larynx; recurrent laryngeal nerve; superior laryngeal nerve.

Introduction

Motor neurons innervating the intrinsic muscles of the larynx are located within the nucleus ambiguus (Amb). The three principal intrinsic laryngeal muscles are supplied by discrete populations of motor neurons. Those innervating the cricothyroid muscle are located most rostrally, in the 'semicompact formation'. An intermediate population innervates the posterior cricoarytenoid muscle and the most caudal of the three populations innervates the thyroarytenoid muscle. The latter two populations have been described to constitute the 'loose formation' (Bieger & Hopkins, 1987; Hernández-Morato et al. 2013a).

The astroglia and microglia of the trigeminal, facial and vagal brainstem nuclei have been shown to react when the their cranial nerves are injured peripherally (Rohlmann et al. 1993, 1994; Ruan et al. 1994; Laskawi & Wolff, 1996; Laskawi et al. 1997; Lan et al. 2000; Storer & Jones, 2003; Hydman et al. 2005; Xu et al. 2008; Lee et al. 2010). However, relatively little information exists about the reactivity of glial cells in the Amb after laryngeal nerve injury. Hydman et al. (2005) reported that resection of the recurrent laryngeal nerve (RLN) resulted in an increased glial fibrillary acidic protein (GFAP) expression in the Amb, which declined slightly by 28 days post-injury (dpi), but nothing is known about the astroglial reaction beyond 28 dpi. Neither is anything known about changes in astroglial reactivity after

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lesions of the superior laryngeal nerve (SLN), or after nerve regeneration. Therefore, the aim of the present study was to investigate the reactivity of astroglia within the Amb, together with glial proliferation and neuronal apoptosis, after denervation and regeneration of the recurrent and superior laryngeal nerves, in an attempt to establish a time frame during which potential therapeutic measures may be applied effectively to modulate the astroglial response to injury.

Materials and methods

Animals

Forty-five adult male Sprague–Dawley rats (*Rattus norvegicus)* of 250–350 g body weight were used in this study (Table 1). They were maintained in the central animal facilities of Complutense University of Madrid and surgery was performed in its animal operating theatre. The research was undertaken in accordance with the laws of Spain (Royal Decree 1201/2005) and the European Union (2010/63/EU) for the care and handling of animals in research and was approved by the Committee of Animal Experimentation of Complutense University. Three of the 45 rats were used as controls for immunohistochemistry procedures.

Surgical procedure: RLN injury and nerve tracing

The rats were anaesthetised with an intra-peritoneal injection of xylazine (8 mg kg^{-1}; Rompun, Bayer, Spain) plus ketamine (90 mg kg^{-1}; Imalgene, Merial, France). The larynx was then exposed through a ventral midline incision in the neck and the skin, salivary glands and infrahyoid muscles were reflected laterally, in order to reveal the left RLN. The left RLN was then isolated and subjected to either resection of 1 mm (n = 18), or transection or the nerve with immediate repair (n = 19) using surgical fibrin glue (Tissucol Duo, Baxter, Spain). The latter group was treated as described for the RLN regeneration model of Hernández-Morato et al. (2014a). For both experimental groups, the RLN was divided at the level of the 6th tracheal cartilage. During the first 2 days following surgery, the animals were treated with a standard analgesic protocol consisting of a dose of buprenorphine (0.05 mg kg^{-1}; Buprex, Reckitt Benckiser Healthcare, UK) plus meloxicam (1.0 mg kg^{-1}; Metacam, Boehringer Ingelheim, Spain) administered every 8 h. The rats were each allowed to survive for one of the following time periods: 3, 7, 14, 28 or 56 days post-injury (dpi; Table 1).

Table 1 The number of rats per experimental group.

	3 dpi	7 dpi	14 dpi	28 dpi	56 dpi
RLN resection	4	5 (3 + 2 BDA)	3	3	3
RLN transection and repair	3	4	4	5	3
SLN resection	–	5 (3 + 2 BDA)	–	–	–

BDA, biotinylated dextran amine; dpi, days post injury; RLN, recurrent laryngeal nerve; SLN, superior laryngeal nerve.

Two rats subjected to RLN transection with repair, received the retrograde tracer biotinylated dextran-amines (BDA; 3 kDa; Molecular Probes, Eugene, OR, USA) applied to the proximal lesioned stump of the nerve before repair as previously described (Pascual-Font et al. 2011), to determine the origin of motor neurons projecting into the nerve. Briefly, immediately after transection, lyophilised BDA powder was applied to the sharp tip of an entomological pin (< 0.5 mm in diameter) and subjected to distilled water vapour. This process was repeated three or four times to recrystallise sufficient tracer. The crystallised BDA was then applied to the proximal stump of the transected nerve. To prevent spread of the tracer to the surrounding structures, a piece of parafilm was placed beneath the isolated nerve. The two rats were sacrificed at 7 dpi.

Surgical procedure: SLN injury and nerve tracing

In a third experimental group (n = 5), the left SLN was located and isolated. Just prior to the division into its external and internal branches, 1 mm of nerve was resected. In two rats, BDA was applied to the proximal stump of the SLN, as described previously (Pascual-Font et al. 2011). The rats were killed at 7 dpi.

Histological procedures

At the end of its allotted survival period, each rat was given a lethal dose of pentobarbital (200 mg kg^{-1}, i.p.; Dolethal, Vetoquinol, France) and perfused through the left cardiac ventricle with 250 mL of 4% paraformaldehyde in 0.1 M phosphate buffer (PB) at pH 7.4 and 4 °C. The brainstem was then removed and immersed in similar fixative solution at 4 °C overnight. It was subsequently cryoprotected in 30% w/v sucrose in PB until it sank, embedded in Tissue-Tek® OCT™ Compound (Sakura Finetek, Alphen aan den Rijn, The Netherlands) and 40-μm frozen sections cut through the medulla oblongata perpendicular to its long axis. The sections were collected in PB in 24-well plates and processed for free-floating immunohistochemistry using primary polyclonal antibodies that label motor neurons (ChAT, goat; 1 : 50, Millipore AB144P; Merck Millipore, Darmstadt, Germany), reactive astrocytes (GFAP, mouse; 1 : 200, Millipore MAB360, Merck Millipore), microglia (Iba-1, rabbit; 1 : 200, Wako 019-19741, Wako Pure Chemical Industries, Ltd, Osaka, Japan), apoptosis (caspase 3 active form, rabbit; 1 : 50, Millipore AB3623, Merck Millipore) and cell proliferation (Ki-67, rabbit; 1 : 300, Millipore AB9260, Merck Millipore). The sections were incubated for 1 h at 4 °C in blocking solution [0.05 M Tris phosphate buffer (TBS) at pH 7.4 with 0.3% Triton X-100, 3% BSA and 10% serum from the species in which the secondary antibody was raised]. The primary antibody was diluted in TBS containing 0.3% Triton X-100, 5% serum and applied to the sections for 48 h at 4 °C. In each immunohistochemistry run, one section was incubated without primary antibody, as a negative control. The sections were then washed several times with TBS containing 0.3% Triton X-100. The secondary antibodies (IgG anti-goat Alexa Fluor 488 conjugated, Molecular Probes A11078; IgG anti-mouse Cy3 conjugated, Millipore AP192C, Merck Millipore; IgG anti-rabbit Alexa Fluor 568 conjugated, Molecular Probes A11011) were diluted 1 : 200 in TBS with 0.3% Triton X-100 and applied to the sections for 24 h at 4 °C in the dark. Finally, after several washes with TBS containing 0.3% Triton X-100, the sections were incubated with DAPI (1 : 1000 in TBS containing 0.3% Triton X-100) for 10 min. The sections were

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placed onto poly-L-lysine-coated slides, dehydrated through a series of alcohols, mounted in DPX (Sigma Aldrich, St. Louis, MO, USA) and coverslipped. Sections from rats killed at different dpi and control rats were processed in parallel.

As proliferation positive controls, H1299, HeLA and HT1080 cells were seeded onto coverslips, grown, fixed and processed for immunohistochemstry in the same way as the sectioned brainstems. As an apoptosis positive control, the growing cell lines described above were treated with different antibiotics (Blasticidin 5 μg mL^{-1}, Zeocin 200 μg mL^{-1}, Puromycin 1. 25 μg mL^{-1}; Gibco, Invitrogen, CA, USA) fixed and treated in the same way as the medulla oblongata sections. To detect the BDA tracer, fluorescein conjugated streptavidin (Vector Laboratories, Peterborough, UK) was diluted 1 : 200 and applied with the secondary antibody.

Image analysis and statistical tests

For each of the five antibodies employed, immunolabelling was performed using 14 sections throughout the length of the medulla oblongata, each separated by 240 μm. The first of each of the 14 series of five serially immunolabelled sections began at 720 μm caudal to the obex and ended 2440 μm rostral to it. The sections were viewed in a Nikon Eclipse 800 M fluorescence microscope equipped with a Nikon DMX1200 (12 megapixel) digital camera. Images of the medulla oblongata were taken at ×200 magnification with the same exposure for each fluorophore quantified, and then the images were subjected to a grey scale transformation and analyzed to quantify the fluorescent signal using IMAGE J software (NIH). The threshold level was determined and then the level of immunoreactivity was assessed by calculating the integrated signal density (mean density above the threshold minus the background) of the region of interest (0.16 mm^2) within the Amb localised by means of ChAT immunohistochemistry. The GFAP immunoreactivity observed in the Amb of the injured (left) side was compared with that of the un-injured side, establishing a ratio, as has been previously described (Laskawi & Wolff, 1996). A ratio of left vs. right GFAP immunoreactivity was also established for control (non-operated) rats (n = 3). For each rat, a ratio was established for each section through the entire extent of the rostrocaudal axis of the Amb. The mean ratio was then calculated as the mean of these values for all the rats of each group. All values are presented as mean ± standard error of the mean (SEM). Statistical comparisons between means were made by analysis of variance (ANOVA) followed by Bonferroni multiple-comparison tests. Differences were considered statistically significant at $P < 0.05$.

Results

GFAP immunoreactivity after RLN injury

Application of the retrograde tracer BDA to the proximal stump of the transected RLN before repair, confirmed the location of RLN motor neuron somata located in the 'loose formation' of the Amb. GFAP immunoreactivity within the region of the Amb containing motor neuron somata projecting into the RLN was significantly increased from 3 dpi ($P < 0.05$) in the resection group and from 7 dpi ($P < 0.001$) in the transection with repair group (Fig. 1). In the resection group, GFAP immunoreactivity remained significantly higher than controls at 7, 14 and 28 dpi ($P < 0.01$ for all times), but thereafter decreased so that at 56 dpi, it was not significantly different from controls (Figs 1 and 2). However, in transection with repair rats, GFAP immunoreactivity remained significantly higher ($P < 0.001$) than controls at 14 dpi, but thereafter declined so that from 28 dpi it was not significantly different from that of controls (Fig. 1). To topographically locate the changes in GFAP immunoreactivity, the obex and other previously established criteria (landmarks such as the caudal-most extent of the dorsal cochlear nucleus and the rostral limit of the area postrema) were used as references and the changes were plotted along the rostrocaudal axis of the Amb (Pascual-Font et al. 2011; Hernández-Morato et al. 2013a); for the neuronal somata contributing to the RLN: –900 to 1680 μm and for the SLN: 1440 to 2590 μm in relation to the obex.

The increase in GFAP immunoreactivity following RLN injury extended beyond the region of the Amb containing RLN motor neuron somata; increased GFAP immunoreactivity was also observed within the region containing SLN motor neuron somata (Fig. 3). After both resection of the RLN and transection with repair, GFAP immunoreactivity was observed 2160 μm rostral to the obex, which corresponds to the location of SLN motor neuron somata (Fig. 4).

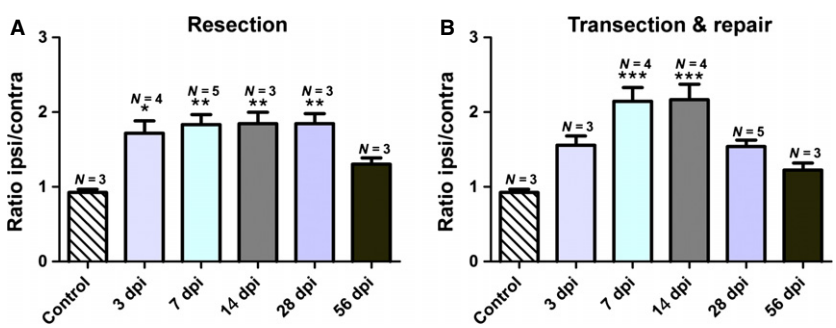


Fig. 1 Bar diagrams showing changes in mean (± SEM) glial fibrillary acidic protein (GFAP) immunofluorescence 'integrated density' (the ratio of the ipsilateral compared to the contralateral immunofluorescence density) throughout the entire extent of the nucleus ambiguus, following recurrent laryngeal nerve resection (A) or transection & repair (B). contra, contralaleral; ipsi, ipsilateral. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$. N = number of rats for each data point.

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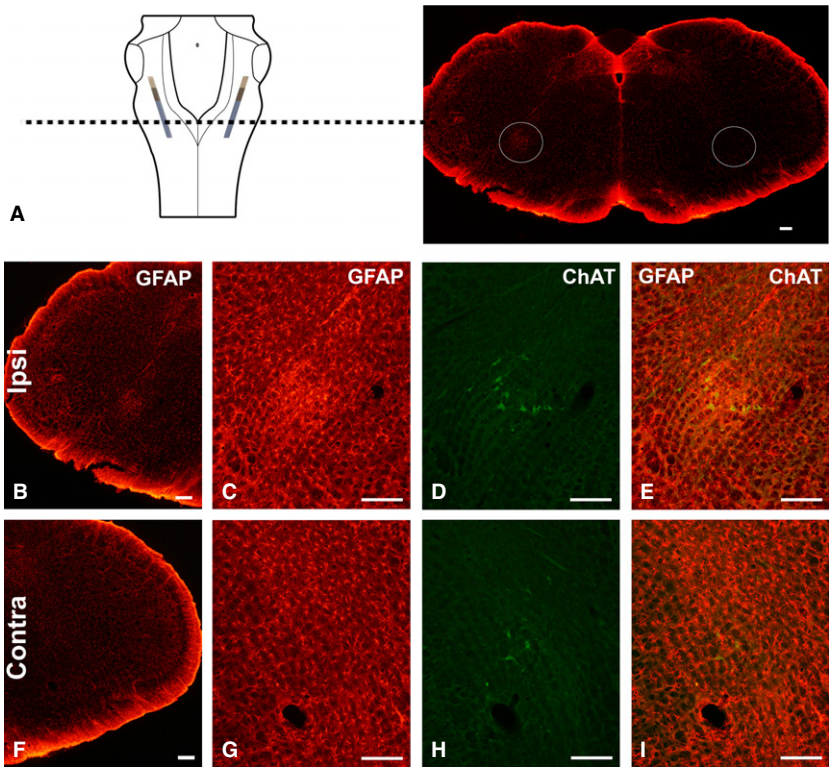


Fig. 2 Immunoreactivity within the region of the nucleus ambiguus (Amb) containing recurrent laryngeal nerve (RLN) motor neuron somata (approximately 240 μm caudal to the obex), 14 days after RLN resection. (A, left) Schematic representation of the medulla oblongata viewed from its dorsal aspect, showing the location of RLN motor neuron somata (grey), superior laryngeal nerve motor neuron somata (light brown) and the rostro-caudal overlap between the two populations (dark brown). The dashed line indicates the level of the medulla oblongata from which the photomicrographs were taken. (A, right) Photomicrograph of a section of the medulla oblongata immunolabelled for glial fibrillary acidic protein (GFAP); the white circles (each 0.16 mm^2) delineate the region of the nucleus ambiguous containing RLN motor neuron somata identified by choline acetyltransferase (ChAT) immunohistochemistry. (B-E) Photomicrographs of the Amb ipsilateral to the RLN resection immunolabelled for GFAP (B,C), ChAT (D), and both GFAP and ChAT (E). (F-I) Photomicrographs of the Amb contralateral to the RLN resection immunolabelled for GFAP (F,G), ChAT (H), and both GFAP and ChAT (I). GFAP imunoreactivity was increased in the Amb ipsilateral to the RLN lesion compared with the contralateral Amb. Scale bars: 200 μm.

There was no increased GFAP immunoreactivity detected in the region between the RLN and SLN motor neuron somata at 1440 μm (Fig. 5). After transection and repair, the levels of the GFAP immunoreactivity gradually decreased after 14 dpi both in the region containing RLN motor neuron somata and in that containing SLN motor neuron somata. In contrast, after RLN resection, the level of GFAP immunoreactivity in the region containing RLN motor neuron somata had decreased to control levels at 56 dpi.

GFAP immunoreactivity after SLN injury

Application of BDA to the proximal stump of the resected SLN confirmed the location of SLN motor neuron somata in the 'semicompact formation' of the Amb. Resection of the SLN resulted in increased GFAP immunoreactivity within the region of the Amb containing SLN motor neuron somata at 7 dpi (Fig. 6). GFAP immunoreactivity was also observed within the region containing motor neuron somata projecting into the RLN (Fig. 7). In contrast to the topographical distribution of GFAP immunoreactivity observed after RLN injury, the increase in GFAP immunoreactivity after SLN resection was continuous between the regions containing SLN and RLN motor neuron somata.

Cell proliferation and apoptosis

H1299, HeLA and HT1080 cells were immunolabelled with the Ki-67 antibody, indicating that they were actively proliferating. In contrast, H1299, HeLA and HT1080 cells treated with antibiotics were immunolabelled with caspase-3 antibody, indicating that they were undergoing apoptosis. However, neither GFAP immunoreactive cells (presumed to be astrocytes) nor Iba-1 immunoreactive cells (presumed to be microglia) in the Amb were immunolabelled with Ki-67. Furthermore, no ChAT immunoreactive cells (presumed to be motor neurons) in the Amb were labelled with the caspase-3 antibody. Therefore, neither glial proliferation nor neuronal apoptosis appear to occur in the Amb after RLN or SLN injury within the timeframe of the current experiments.

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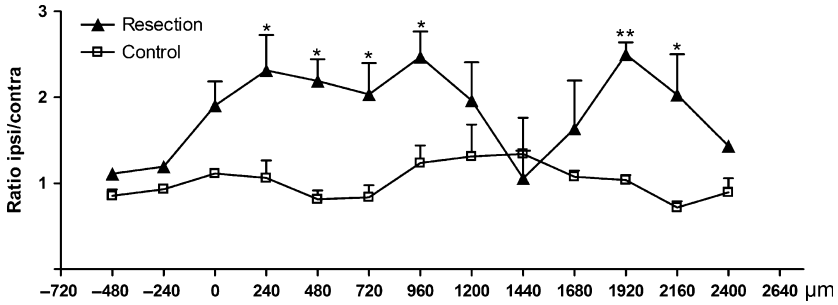


Fig. 3 Graph showing rostro-caudal differences in glial fibrillary acidic protein (GFAP) immunoreactivity in the nucleus ambiguus 7 days after recurrent laryngeal nerve (RLN) resection compared with uninjured contralateral control nerves. The 'integrated density' of GFAP immunofluorescence (the ratio of the ipsilateral compared to the contralateral immunofluorescence density) is plotted against the distance from the obex (0). Significantly increased GFAP immunoreactivity was observed in the nucleus ambiguus on the side of the resected RLN not only in the region containing RLN motor neuron somata (−900 to 1800 µm), but also in the region containing superior laryngeal motor neuron somata (1230 to 2590 µm). contra, contralaleral; ipsi, ipsilateral. * $P < 0.05$; ** $P < 0.01$. $n = 5$ for each data point.

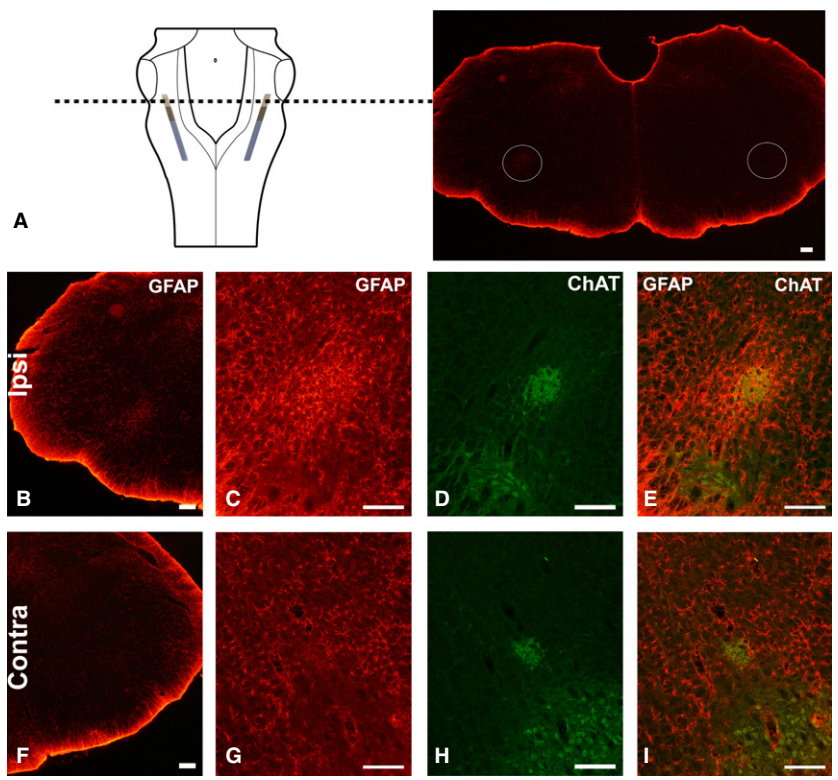


Fig. 4 Immunoreactivity within the region of the nucleus ambiguus (Amb) containing superior laryngeal nerve (SLN) motor neuron somata (approxmately 2160 µm rostral to the obex), 14 days after recurrent laryngeal nerve (RLN) resection. (A, left) Schematic representation of the medulla oblongata viewed from its dorsal aspect, showing the location of RLN motor neuron somata (grey), superior laryngeal nerve motor neuron somata (light brown) and the rostro-caudal overlap between the two populations (dark brown). The dashed line indicates the level of the medulla oblongata from which the photomicrographs were taken. (A, right) Photomicrograph of a section of the medulla oblongata immunolabelled for glial fibrillary acidic protein (GFAP); the white circles (each 0.16 mm^2) delineate the region of the nucleus ambiguus containing SLN motor neuron somata identified by choline acetyltransferase (ChAT) immunohistochemistry. (B-E) Photomicrographs of the Amb ipsilateral to the RLN resection immunolabelled for GFAP (B,C), ChAT (D), and both GFAP and ChAT (E). (F-I) Photomicrographs of the Amb contralateral to the RLN resection immunolabelled for GFAP (F,G), ChAT (H), and both GFAP and ChAT (I). GFAP imunoreactivity was increased in the Amb ipsilateral to the RLN lesion compared with the contralateral Amb. ChAT immunolabelling (D,H) reveals motor neurons in the 'semicompact formation' of the Amb and strong immunolabelling of large motor neurons in the facial nucleus ventrolaterally. Scale bars: 200 µm.

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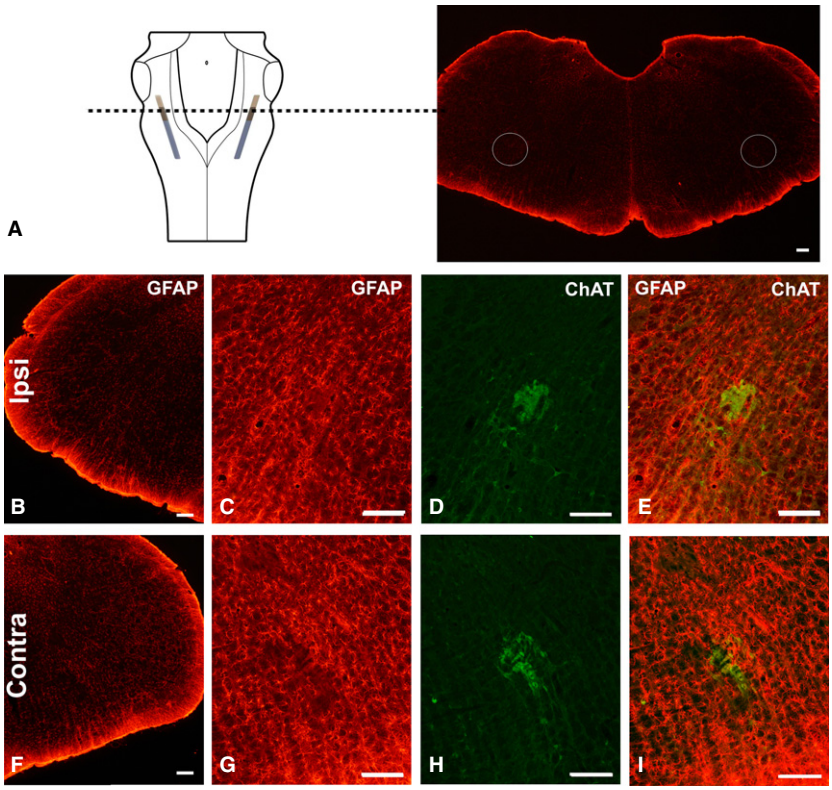


Fig. 5 Immunoreactivity within the region of the nucleus ambiguus (Amb) containing overlapping populations of recurrent laryngeal nerve (RLN) and superior laryngeal nerve (SLN) motor neuron somata (approximately1440 μm rostral to the obex), 14 days after RLN transection with repair section. (A, left) Schematic representation of the medulla oblongata viewed from its dorsal aspect, showing the location of RLN motor neuron somata (grey), SLN motor neuron somata (light brown) and the rostro-caudal overlap between the two populations (dark brown). The dashed line indicates the level of the medulla oblongata from which the photomicrographs were taken. (A, right) Photomicrograph of a section of the medulla oblongata immunolabelled for glial fibrillary acidic protein (GFAP); the white circles (each 0.16 mm^2) delineate the region of the nucleus ambiguous containing both RLN and SLN motor neuron somata identified by choline acetyltransferase (ChAT) immunohistochemistry. (B-E) Photomicrographs of the Amb ipsilateral to the RLN transection with repair, immunolabelled for GFAP (B,C), ChAT (D), and both GFAP and ChAT (E). (F-I) photomicrographs of the Amb contralateral to the RLN resection immunolabelled for GFAP (F,G), ChAT (H), and both GFAP and ChAT (I). GFAP imunoreactivity was not significantly different in the Amb ipsilateral to the RLN transection with repair, compared with the contralateral Amb. Scale bars: 200 μm.

Discussion

The increased GFAP immunoreactivity (reactive astrocytes) observed after RLN resection and transection with repair is similar to the increased GFAP reactivity at 7 dpi reported by Hydman et al. (2005) following 2-mm resection of the RLN. In addition, the results of the current study reveal that increased GFAP immunoreactivity is already present at 3 dpi following resection of the RLN. Furthermore, GFAP immunoreactivity was observed in the Amb, 7 dpi to the SLN, a nerve that has not previously been investigated. GFAP immunoreactivity returned to basal levels between 14 and 28 dpi following transection and repair, compared with between 28 and 56 dpi following resection of the RLN. The decrease in GFAP immunoreactivity observed in the Amb in the current study at 28 dpi following RLN transection with repair supports the earlier finding that after transection and repair of the RLN, axonal transport recovered between 28 and 56 dpi (Hernández-Morato et al. 2014a). These results indicate when the axon reaction ceases in the Amb and when the regenerating axons reach their muscle targets. Axonal transport has been shown to take longer to recover after RLN resection than after transection and repair (Pascual-Font et al. 2008). This finding is in agreement with the finding in the current study that, following RLN resection, GFAP immunoreactivity levels remain elevated for longer than following transection with repair. Similar results have been reported following facial nerve injury, when elevated GFAP immunoreactivity in the facial nerve nucleus can remain for several months if axonal regeneration is impeded, but recovery time is much shorter if the nerve is repaired (Guntinas-Lichius et al. 1994; Laskawi & Wolff, 1996).

The results of the current study are the first precisely to identify the region of increased GFAP immunoreactivity throughout the entire rostro-caudal extent of the Amb fol-

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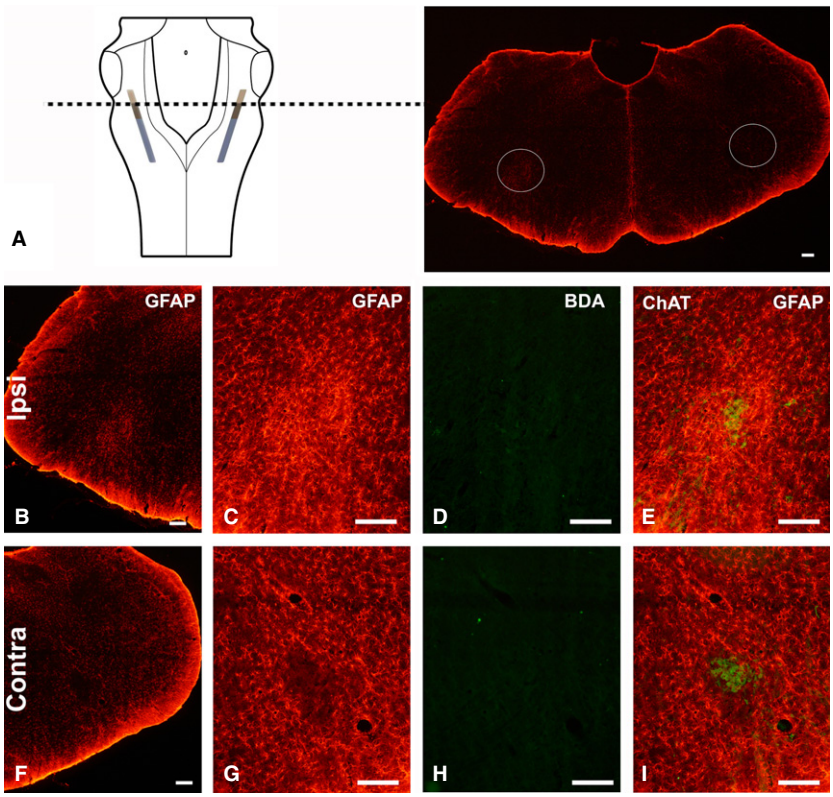


Fig. 6 Immunoreactivity within the region of the nucleus ambiguus (Amb) containing superior laryngeal nerve (SLN) motor neuron somata (approximately 1920 μm rostral to the obex), 7 days after SLN resection. (A, left) Schematic representation of the medulla oblongata viewed from its dorsal aspect, showing the location of recurrent laryngeal nerve motor neuron somata (grey), SLN motor neuron somata (light brown) and the rostro-caudal overlap between the two populations (dark brown). The dashed line indicates the level of the medulla oblongata from which the photomicrographs were taken. (A, right) Photomicrograph of a section of the medulla oblongata immunolabelled for glial fibrillary acidic protein (GFAP); the white circles (each 0.16 mm^2) delineate the region of the nucleus ambiguous containing SLN motor neuron somata identified by choline acetyltransferase (ChAT) immunohistochemistry. (B-E) Photomicrographs of the Amb ipsilateral to the SLN resection immunolabelled for GFAP (B,C), neuronal somata labelled with biotinylated dextran amines (BDA) applied to the proximal stump of the resected SLN (D) and both GFAP and ChAT immunolabeling combined (E). (F-I) Photomicrographs of the Amb contralateral to the SLN resection immunolabelled for GFAP (F,G), the contralateral immunofluorescence control (H) for the BDA labelling shown in D, and both GFAP and ChAT immunolabelling combined (I). GFAP immunoreactivity was increased in the Amb ipsilateral to the SLN lesion compared with the contralateral Amb. Scale bars: 200 μm.

lowing RLN injury, revealing that the increased GFAP reactivity extended beyond the region containing the motor neuron somata of the injured nerve. When the RLN was injured, increased GFAP reactivity was also observed in the region containing SLN motor neuron somata (Bieger & Hopkins, 1987; Pascual-Font et al. 2011) and when the SLN was injured, increased GFAP immunoreactivity was also found in the region containing RLN motor neuron somata (Bieger & Hopkins, 1987; Pascual-Font et al. 2011). However, unlike after SLN injury, following RLN resection or transection and repair, the increased GFAP immunoreactivity was not continuous along the rostro-caudal extent of the Amb between the regions containing RLN and SLN motor neuron somata. Therefore, the GFAP immunoreactivity in the region containing SLN motor neuron somata is unlikely to have been caused by factors diffusing from the region of RLN motor neuron somata or through signalling across gap junctions between astrocytes. It is possible that the regions containing RLN and SLN motor neuron somata are connected by interneurons; Amb motor neurons are connected through the reticular formation of the medulla in order to control respiratory oscillations and synchronised glottic movements (Yajima & Hayashi, 1997), but this would be unlikely to explain increased GFAP immunoreactivity in discrete regions of the nucleus rather than in the Amb as a whole. It is possible that after RLN lesion, the SLN sprouts axon collaterals to muscles normally innervated by the RLN; thus the increased GFAP immunoreactivity could reflect a regenerative rather than a degenerative response. Such a phenomenon would also explain why, after a RLN crush injury, retrograde tracers applied to RLN innervated muscles label motor neuron somata in the region of the Amb that gives rise to the SLN and innervates the posterior cricoarytenoid muscle (Hernández-Morato et al. 2013b).

A plausible explanation for the increased GFAP immunoreactivity in both the regions of the Amb containing motor

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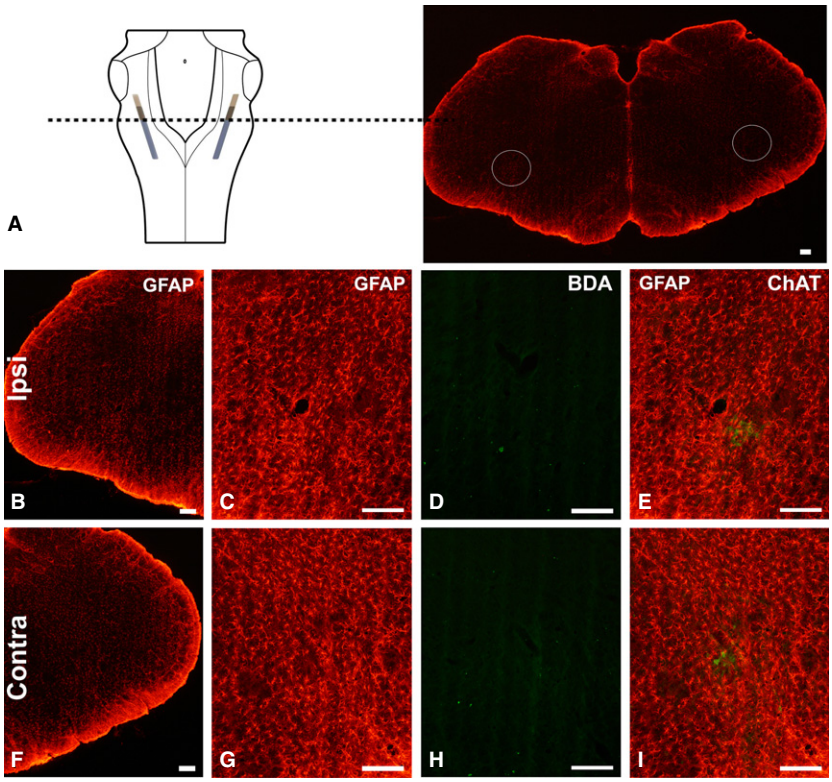


Fig. 7 Immunoreactivity within the region of the nucleus ambiguus (Amb) containing recurrent laryngeal nerve (RLN) motor neuron somata (approximately 1200 μm rostral to the obex), 7 days after superior laryngeal nerve (SLN) resection. (A, left) Schematic representation of the medulla oblongata viewed from its dorsal aspect, showing the location of RLN motor neuron somata (grey), SLN motor neuron somata (light brown) and the rostro-caudal overlap between the two populations (dark brown). The dashed line indicates the level of the medulla oblongata from which the photomicrographs were taken. (A, right) Photomicrograph of a section of the medulla oblongata immunolabelled for glial fibrillary acidic protein (GFAP); the white circles (each 0.16 mm^2) delineate the region of the nucleus ambiguous containing RLN motor neuron somata identified by choline acetyltransferase (ChAT) immunohistochemistry. (B-E) Photomicrographs of the Amb ipsilateral to the SLN resection immunolabelled for GFAP (B,C), following application of biotinylated dextran amines (BDA) to the proximal stump of the resected SLN (D), and both GFAP and ChAT immunolabelling combined (E). (F-I) Photomicrographs of the Amb contralateral to the SLN resection immunolabelled for GFAP (F,G), the contralateral immunofluorescence control (H) for application of BDA to the resected SLN and both GFAP and ChAT immunolabelling combined (I). GFAP immunoreactivity increased in the Amb ipsilateral to the SLN resection compared with the contralateral side and application of BDA to the resected SLN did not result in labelling of neuronal somata in the Amb, confirming that this region of the Amb only contains RLN motor neuron somata. Scale bars: 200 μm.

neuron somata projecting into the RLN and the SLN, following RLN injury, lies in the observation that the posterior cricoarytenoid muscle receives a dual innervation from the RLN and SLN (Hydman & Mattsson, 2008). Thus, after RLN injury, astrocytes surrounding axotomised RLN motor neuron somata would become reactive, expressing GFAP, as would those astrocytes around motor neuron somata that innervate the posterior cricoarytenoid muscle via the SLN. The region of the Amb between the two regions of increased GFAP immunoreactivity remains GFAP-immunonegative after RLN injury because it is the region of the Amb that innervates the cricothyroid muscle through the SLN alone. This explanation is supported by the fact that, after SLN injury, continuous increased GFAP immunoreactivity was observed through the regions of the Amb containing RLN and SLN motor neuron somata, which are likely to innervate the cricothyroid and posterior cricoarytenoid muscles. These motor neuron somata are located rostrally in the 'loose' and the 'semicompact' formations of the Amb, respectively (Hernández-Morato et al. 2013a).

The fact that, following RLN injury, GFAP immunoreactivity in the region of the Amb containing motor neuron somata projecting into the SLN is not as great as it is in the region containing RLN motor neuron somata can be explained by the fact that only 25% of the axons supplying the posterior cricoarytenoid muscle travel in the SLN, whereas 75% travel in the RLN (Hydman & Mattsson, 2008). Therefore, the amount of compensatory activity performed by uninjured SLN motoneurons in cases of RLN injury should be larger than that of RLN compensation after SLN injury. This compensation could include intramuscular sprouting, which would mean additional metabolic stress on the motor neuron somata, resulting in local astrogliosis. Reactive astrocytes could contribute to recovery after nerve

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injury by supporting neuronal growth and metabolism (reviewed by Angelov & Neiss, 1994).

The dual innervation of a muscle by motor neurons with somata located in different regions of the brainstem may provide a useful model for the study of the mechanisms controlling the astroglial reaction to axonal injury. For example, it may be possible to study the signals sent by the posterior cricoarytenoid muscle along an intact nerve after denervation of its other nerve supply. It may be possible to inhibit or enhance these signals that travel along the intact nerve and to study how astroglia respond to them. It may also be possible to compare these signals with those responsible for the astroglial activation in the region containing the motor neuron somata of the injured nerve and to describe two types of astroglial reaction.

It is important to note that the reaction of the Amb to laryngeal nerve lesions shown in this and the study of Hydman et al. (2005) is different to the reaction to injury to other nerves, such as the facial nerve, where proliferation and activation of microglia or neuronal apoptosis have been observed (Graeber et al. 1988; Streit et al. 1988). This difference may be due to the proximity of the nerve lesion to the motor neuron somata. It has been reported that, even in the same model system, proximity of the nerve injury to the motor neuron somata affects a number of parameters of the glial and neuronal response (Liu et al. 2006). In the laryngeal nerve injury model (both RLN and SLN) the nerve is damaged at a very distal point, near to its entry to the muscle it supplies. This could explain the lack of central apoptosis, astroglial or microglia proliferation following any type of laryngeal nerve injury employed in the current study.

A possible hypothesis to explain the results of the current study is that, after lesion of the RLN, the partially denervated posterior cricoarytenoid muscle sends a signal that travels along the intact SLN nerve to the SLN motor neuron somata in the 'semicompact formation' of the Amb, which respond by activating axonal sprouting within the posterior cricoarytenoid muscle. In the postnatal and adult rat, GDNF (glial-derived neurotrophic factor) is continuously produced by skeletal muscle and is transported retrogradely to the somata of neurons innervating them (Yan et al. 1995; Russell et al. 2000). After sciatic nerve injury, the levels of GDNF and IGF-II are upregulated in the gastrocnemius muscle in the rat and in the soleus and gastrocnemius in the mouse (Glazner & Ishii, 1995; Naveilhan et al. 1997). GDNF and other neurotrophic factors (brain-derived neurotrophic factor, Neurotrophine 4) are upregulated in the laryngeal muscles after injury to the recurrent laryngeal nerve (Vega-Cordova et al. 2010; Hernández-Morato et al. 2014b). Therefore, it is possible that following RLN injury, the partially dennervated posterior cricoarytenoid muscle responds by upregulating GDNF or other trophic factors that are transported retrogradely to the 'semicompact formation' of the Amb by the intact SLN. This process would cause additional metabolic stress, resulting in increased GFAP immunoreactivity in the region of the SLN motor neuron somata. Similarly, after SLN injury, the partially denervated posterior cricoarytenoid muscle could send a signal along the intact RLN. This signal would reach the region containing RLN motor neuron somata within the 'loose formation' of the Amb, resulting in increased GFAP immunoreactivity within it and some axonal sprouting within the posterior cricoarytenoid muscle. The increase in GFAP immunoreactivity after SLN nerve injury is smaller than after RLN injury because the posterior cricoarytenoid is less denervated after SLN injury than after RLN injury due to its unequal innervation.

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Conflict of interest

The authors declare that they have no conflict of interest.

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